

1.1 Matrix Operations

Consider the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 1.3443 & -2.3342 & 4.5461 \\ -2.3234 & 3.4535 & -5.6617 \end{bmatrix}, \quad (1)$$

$$B = \begin{bmatrix} 2.3724 & 4.5212 \\ 3.4564 & 6.7448 \\ 1.2643 & -2.3254 \end{bmatrix}, \quad (2)$$

$$C = \begin{bmatrix} -1.2373 & 1.2334 \\ 3.3122 & 3.2122 \end{bmatrix} \quad (3)$$

Task: Determine, if possible, the following matrix products:

If any computation is impossible, explain why based on matrix dimension compatibility.

Instructions: You may use calculators for computations. You don't need to show the detailed calculation steps. Compute with full precision, but round your final answers to 3 decimal places.

① CA

Since $C \in \mathbb{R}^{2 \times 2}$, $A \in \mathbb{R}^{2 \times 3} \Rightarrow CA \in \mathbb{R}^{2 \times 3}$

$$CA_{ij} = \sum_{k=1}^2 C_{ik} A_{kj}, \quad CA = \begin{bmatrix} -4.529 & 7.148 & -12.608 \\ -3.011 & 3.362 & -3.129 \end{bmatrix}$$

② BC

Since $B \in \mathbb{R}^{3 \times 2}$, $C \in \mathbb{R}^{2 \times 2} \Rightarrow BC \in \mathbb{R}^{3 \times 2}$

$$BC_{ij} = \sum_{k=1}^2 B_{ik} C_{kj}, \quad BC = \begin{bmatrix} 12.040 & 17.449 \\ 18.064 & 25.929 \\ -9.267 & -5.910 \end{bmatrix}$$

③ $B^T A$

$B^T \in \mathbb{R}^{2 \times 3}$, $A \in \mathbb{R}^{2 \times 3} \Rightarrow$ num of columns of $B^T \neq$ num of rows of A
 $\Rightarrow B^T A$ is not defined

④ $A^T C$

Since $A^T \in \mathbb{R}^{3 \times 2}$, $C \in \mathbb{R}^{2 \times 2} \Rightarrow A^T C \in \mathbb{R}^{3 \times 2}$

$$A^T C = \begin{bmatrix} -9.359 & -5.805 \\ 14.327 & 8.214 \\ -24.378 & -12.579 \end{bmatrix}$$

⑤ $(-A)^T C$

Since $(-A)^T \in \mathbb{R}^{3 \times 2}$, $C \in \mathbb{R}^{2 \times 2} \Rightarrow (-A)^T C \in \mathbb{R}^{3 \times 2}$

$$(-A^T C) = \begin{bmatrix} 9.359 & 5.805 \\ -14.327 & -8.214 \\ 24.378 & 12.579 \end{bmatrix}$$

⑥ $B^T A^T$

Since $B^T \in \mathbb{R}^{2 \times 3}$, $A^T \in \mathbb{R}^{3 \times 2}$, $B^T A^T \in \mathbb{R}^{2 \times 2}$

$$B^T A^T = \begin{bmatrix} 0.869 & -0.733 \\ -20.237 & 25.954 \end{bmatrix}$$

⑦ AC

$A \in \mathbb{R}^{2 \times 3}$, $C \in \mathbb{R}^{2 \times 2} \Rightarrow$ num of columns of $A \neq$ num of rows of C

$\Rightarrow AC$ is not defined

⑧ CB

$C \in \mathbb{R}^{2 \times 2}$, $B \in \mathbb{R}^{3 \times 2} \Rightarrow$ num of columns of $C \neq$ num of rows of B

$\Rightarrow CB$ is not defined

1.2 Properties of Determinants

Task: Construct matrices $A, B \in \mathbb{R}^{n \times n}$ such that $\det(A+B) \neq \det(A) + \det(B)$ for:

(a) $n = 2$

(b) $n = 3$

(c) $n = 4$

For each case, provide specific matrices and verify your result by computing the determinants.

(d) Can you find a general pattern or construction method that works for any n ?

Instructions: You may use calculators for determinant computations. You don't need to show detailed calculation steps, but your equations and final results must be accurate.

(a) $n=2$

$$\text{Let } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A+B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\det(A) = 1, \det(B) = 1, \det(A+B) = 4$$

$$\text{So, } \det(A+B) \neq \det(A) + \det(B)$$

(b) $n=3$

$$\text{Let } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow A+B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\det(A) = 1, \det(B) = 1, \det(A+B) = 8$$

$$\text{So, } \det(A+B) \neq \det(A) + \det(B)$$

(c) $n=4$

$$\text{Let } A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow A+B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\det(A) = 1, \det(B) = 1, \det(A+B) = 16$$

$$\text{So, } \det(A+B) \neq \det(A) + \det(B)$$

(d)

I found general construction at $n \geq 2$,

$$A = aI_n, \quad B = bI_n \quad (a, b \neq 0)$$

$$A = \begin{bmatrix} a & 0 & \dots & 0 \\ 0 & a & & \\ \vdots & & \ddots & \\ 0 & \dots & 0 & a \end{bmatrix}, \quad B = \begin{bmatrix} b & 0 & \dots & 0 \\ 0 & b & & \\ \vdots & & \ddots & \\ 0 & \dots & 0 & b \end{bmatrix}, \quad A+B = \begin{bmatrix} a+b & 0 & \dots & 0 \\ 0 & a+b & & \\ \vdots & & \ddots & \\ 0 & \dots & 0 & a+b \end{bmatrix}$$

$$\det(A) = a^n, \quad \det(B) = b^n, \quad \det(A+B) = (a+b)^n$$

$$\text{For } \forall n \geq 2, \quad (a+b)^n > a^n + b^n \quad (a, b \neq 0)$$

So, If $A = aI_n, B = bI_n$ with scalars $a, b \neq 0$,

$$\det(A) + \det(B) \neq \det(A+B)$$

1.3 Computing Eigenvalues and Eigenvectors

Task: Determine the **eigenvalues** and corresponding **eigenvectors** of the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad (4)$$

$$B = \begin{bmatrix} 4 & -12 & 5 \\ 0 & 2 & 13 \\ 0 & 3 & -1 \end{bmatrix}, \quad (5)$$

$$C = \begin{bmatrix} 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix} \quad (6)$$

Additional Question: Is there any difference when you consider A , B , and C as matrices over $\mathbb{C}$ instead of $\mathbb{R}$?

① A

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 & 1 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix} \Rightarrow \det(A - \lambda I) = (1-\lambda) \{ (1-\lambda)^2 - 1 \cdot 0 \} \\ = (1-\lambda)^3$$

$$(1-\lambda)^3 = 0 \Rightarrow \underline{\lambda = 1} \text{ is eigenvalue of } A.$$

$$\text{Then, } (A - I)v = 0, \quad v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} v = 0 \quad \left(\begin{array}{l} y+z=0 \\ z=0 \end{array} \right) \Rightarrow y=0, z=0$$

$$\underline{v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} s} \quad (s \neq 0) \text{ is corresponding eigenvectors.}$$

② B

$$B - \lambda I = \begin{bmatrix} 4-\lambda & -12 & 5 \\ 0 & 2-\lambda & 13 \\ 0 & 3 & -1-\lambda \end{bmatrix}$$

$$\Rightarrow \det(B - \lambda I) = (4-\lambda) \{ (2-\lambda)(-1-\lambda) - 39 \}$$
$$= (4-\lambda)(\lambda^2 - \lambda - 41)$$

$$\lambda_1 = 4, \quad \lambda_2 = \frac{1+\sqrt{165}}{2}, \quad \lambda_3 = \frac{1-\sqrt{165}}{2}$$

are eigenvalues of B.

i) $\lambda_1 = 4$

$$B - 4I = \begin{bmatrix} 0 & -12 & 5 \\ 0 & -2 & 13 \\ 0 & 3 & -5 \end{bmatrix} v_1 = 0 \Rightarrow \underline{v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} s} \quad (s \neq 0)$$

ii) λ_2, λ_3 is corresponding eigenvector of λ_1

$$\begin{bmatrix} 4-\lambda & -12 & 5 \\ 0 & 2-\lambda & 13 \\ 0 & 3 & -1-\lambda \end{bmatrix} v = 0, \quad v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$3y + (-1-\lambda)z = 0 \rightarrow y = \frac{\lambda+1}{3}z$$

$$(4-\lambda)x - 12y + 3z = 0 \rightarrow x = \frac{12y - 3z}{4-\lambda} = \frac{4(\lambda+1)z - 5z}{4-\lambda} = \frac{4\lambda-1}{4-\lambda}z$$

($\lambda \neq 4$)

$$\text{Let } z=1, \quad V = \begin{bmatrix} \frac{4\lambda-1}{4-\lambda} \\ \frac{\lambda+1}{3} \\ 1 \end{bmatrix}$$

$$\text{If } \lambda = \lambda_2 = \frac{1+\sqrt{165}}{2},$$

$$V_2 = \begin{bmatrix} -\frac{337}{58} - \frac{15\sqrt{165}}{58} \\ \frac{1}{2} + \frac{\sqrt{165}}{6} \\ 1 \end{bmatrix} s$$

($s \neq 0$)

is corresponding
eigenvector of λ_2 .

$$\text{If } \lambda = \lambda_3 = \frac{1-\sqrt{165}}{2}$$

$$V_3 = \begin{bmatrix} -\frac{337}{58} + \frac{15\sqrt{165}}{58} \\ \frac{1}{2} - \frac{\sqrt{165}}{6} \\ 1 \end{bmatrix} s$$

($s \neq 0$)

is corresponding
eigenvector of λ_3 .

③ C

$$C - \lambda I = \begin{bmatrix} -\lambda & -2 & 0 & 0 \\ 2 & -\lambda & 0 & 0 \\ 0 & 0 & -2-\lambda & 2 \\ 0 & 0 & 0 & -2-\lambda \end{bmatrix}$$

$$\begin{aligned} \det(C - \lambda I) &= (-\lambda)(-\lambda) \cdot \{(2-\lambda)^2 - 2 \cdot 0\} - (-2) \cdot 2 \cdot \{(-2-\lambda)^2 - 2 \cdot 0\} \\ &= (\lambda^2 + 4) (\lambda + 2)^2 \quad (\lambda \in \mathbb{R}) \end{aligned}$$

$\Rightarrow \underline{\lambda = -2}$ is eigenvalue of C

$$C + 2I = \begin{bmatrix} 2 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$V = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} s$ ($s \neq 0$) is corresponding eigenvector.

* Additional question

At A and B, there are only real number eigenvalues.
So, there is no difference between considering them over $\mathbb{R}$ and $\mathbb{C}$.

However, At C, $\lambda = 2i$ or $\lambda = -2i$ can be eigenvalues.

$$\text{Let } v = \begin{bmatrix} x \\ y \\ z \\ d \end{bmatrix}$$

$$\text{By } (C - \lambda I)v = 0, \quad C - \lambda I = \begin{bmatrix} -\lambda & -2 & 0 & 0 \\ 2 & -\lambda & 0 & 0 \\ 0 & 0 & -2-\lambda & 2 \\ 0 & 0 & 0 & -2-\lambda \end{bmatrix}$$

$$(-2-\lambda)d = 0 \rightarrow d = 0$$

$$(2-\lambda)z + 2d = 0 \rightarrow z = 0$$

$$-\lambda x - 2y = 0 \rightarrow y = -\frac{\lambda}{2}x$$

$$\text{i) } \lambda = 2i$$

$$y = -ix$$

$$v = \begin{bmatrix} 1 \\ -i \\ 0 \\ 0 \end{bmatrix} s \quad (s \neq 0)$$

is corresponding eigenvector
of $\lambda = 2i$

$$\text{ii) } \lambda = -2i$$

$$y = ix$$

$$v = \begin{bmatrix} 1 \\ i \\ 0 \\ 0 \end{bmatrix} s \quad (s \neq 0)$$

corresponding eigenvector
of $\lambda = -2i$

Thus, additional eigenvalues and eigenvector can appear over $\mathbb{C}$.